

<< Flood Monitoring & Alerting System>>

**Project Report submitted in partial fulfilment of the requirements for the award of
the degree of**

Bachelor of Technology

In

ELECTRONICS & COMMUNICATION ENGINEERING

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CERTIFICATE

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I am glad to inform that the work is entirely original and its performance is found to be quite satisfactory.

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ABSTRACT

Flooding is considered one of the most destructive natural disasters in the world. In countries like India with climatic conditions occurrence of heavy rain fall and subsequent discharge of water leads to Flood. Flooding creates major damages to life, their habitats and the economy. The loss of properties and living population is getting enhanced every year due to the dynamic alterations in weather conditions which results in heavy floods. Therefore, implementation of an intelligent analysis of flood risk is necessitated for the field of research in Disaster management. Normal Weather monitoring and alerting systems are not quick and accurate enough to predict floods in time to prevent personal or environmental damages. By installing smart flood alerting systems near major waterways vital information can be collected so that lives and property can be protected. The government has to spend tons of money in flood mitigation plans to help the victims and also to reduce the number in the long run damages that can occur after flooding. Nowadays every prediction level has been moved from physical to mathematical modelling. Where, this paper deals with the strategies of flood alerting system with help of an embedded system which would give the real time calculation along with data for computational processing, prediction and analysis that would help to send an alert signal to the nearby surrounding and reduces the time of risk. Here the Arduino uno is connected to water flow sensors to analyse the water level. Further, these analysed values will be passed to the Arduino which is been developed with programming languages like C.

Keywords:

Internet of things, Arduino, Sensors, Modelling.

INTRODUCTION

The extreme climatic changes due to the effect from various human activities such as pollutions, cutting of innumerable trees and too much of gas emission are the some of the main reason for natural disasters that occur in worldwide. The most common factor that cause major damage to life, property and country's economy is the flood. A flood can be described as an overflow of water in an area considered to be relatively dry. They can cause considerable damage to property and the environment, including crops and wildlife, and are an additional financial cost. Floods can be caused by natural disasters or water pipe breaks. Flooding is also associated with the development of mold which can appear within 24 hours. Overflowing rivers can lead to flooding the areas along the river. If heavy precipitation is received upstream of a river, the large water volumes can flood the houses of people who live along the river downstream, even if it did not rain severely there. Broken dams can also lead to flooding. Aging dams, which can be overwhelmed by high levels of water, can fail and unleash torrents of water to the unsuspecting residents, causing devastating flooding. Melting snow and ice after a heavy snowfall and other forms of precipitation can lead to flooding once temperatures begin to warm. Because of the devastating effects that floods can have on people and their environments, flood monitoring systems have been developed to help prepare and warn people of emanating danger. While some areas are more prone to flooding than others, the establishment of flood warning systems near any major waterway or body of water provides critical information that can protect property and save lives. Of course, the most effective flood warning methods extend beyond the installation of gages and telemetry equipment, and employ qualified staff and carefully designed procedures to provide the earliest warning about whether a flood should be expected, when it will occur, and how severe it will be.

This guide offers instruction to individuals, communities, and organizations interested in establishing and operating floodwarning systems. The systems can help prevent excessive damage and loss as a result of flooding and possibly save lives. To manage floods, alerting people, understanding of increased water level and speed of water flow are valuable for discovering potential seriousness of the flood. This project presents the details of how the data - like flood level and rain intensity are collected from sensors and made available on cloud and sending alert messages by using Bolt IOT module and Arduino Uno and a Global System for Mobile communication (GSM) and short message service (SMS) to relay data from sensors to computers or directly alert the people present in that area through their mobile phone. The data from the IOT cloud can be accessed by android smart phones at anytime from anywhere in the world using the mobile app things view.

Some benefits of using a flood monitoring system include:

- High reliability as data is sent in real-time.
- Timely detection of flood risks.
- Tailored solutions that can easily be integrated with external development at any level such as connectivity, user application, and device.

Data collected using a flood monitoring system can be used in future studies for weather patterns and climate change. Many western European countries were recently subjected to severe flooding which resulted in huge economic losses and multiple deaths. Floods have become a frequent threat as a result of climate change, but with the development of artificial intelligence (AI), a much more advanced flood detection system has been developed based on deep computer algorithms. The development of smart cameras and sensor technology helps detect the water levels and measure the probability of flooding even before the floods happen. With much more research being carried out to improve flood monitoring systems, many governments are investing in much quicker methods of detection. Aside from

Real-time analysis, visualization will also help emergency authorities to showcase current conditions, allowing citizens to take action and prepare adequately. Flood monitoring systems also have total integration and adaptation with emergency plans. They consume less energy and can be powered with solar energy. Flood monitoring system equipment is highly resilient and has a long working life span, making it very convenient and cost- effective.

PROBLEMS FACED DUE TO FLOODS

Floods impact on both individuals and communities, and have social, economic, and environmental consequences. Flooding of areas used for socio-economic activities produces a variety of negative impacts. The magnitude of adverse impacts depends on the vulnerability of the activities and population and the frequency, intensity and extent of flooding. Some of these factors are shown below:

Loss of lives and property: Immediate impacts of flooding include loss of human life, damage to property, destruction of crops, loss of livestock, non-functioning of infrastructure facilities and deterioration of health condition owing to waterborne diseases. Flash floods, with little or no warning time, cause more deaths than slow-rising riverine floods.

Loss of livelihoods: As communication links and infrastructure such as power plants, roads and bridges are damaged and disrupted, economic activities come to a standstill, resulting in dislocation and the dysfunction of normal life for a period much beyond the duration of the flooding. Similarly, the direct effect on production assets, be it in agriculture or industry, can inhibit regular activity and lead to loss of livelihoods. The spill over effects of the loss of livelihoods can be felt in business and commercial activities even in adjacent non-flooded areas.

Decreased purchasing and production power: Damage to infrastructure also causes long-term impacts, such as disruptions to clean water and electricity, transport, communication, education and health care. Loss of livelihoods, reduction in purchasing power and loss of land value in the flood plains lead to increased vulnerabilities of communities living in the area. The additional cost of rehabilitation, relocation of people and removal of property from flood-affected areas can divert the capital required for maintaining production.

Mass migration: Frequent flooding, resulting in loss of livelihoods, production and other prolonged economic impacts and types of suffering can trigger mass migration or population displacement.

Migration to developed urban areas contributes to the overcrowding in the cities. These migrants swell the ranks of the urban poor and end up living in marginal lands in cities that are prone to floods or other risks. Selective out-migration of the workforce sometimes creates complex social problems.

Psychosocial effects: The huge psycho-social effects on flood victims and their families can traumatize them for long periods of time. The loss of loved ones can generate deep impacts, especially on children.

Displacement from one's home, loss of property and livelihoods and disruption to business and social affairs can cause continuing stress. The stress of overcoming these losses can be overwhelming and produce lasting psychological impacts.

Hindering economic growth and development: The high cost of relief and recovery may adversely impact investment in infrastructure and other development activities in the area and in certain cases may cripple the frail economy of the region. Recurrent flooding in a region may discourage long-term investments by the government and private sector alike. Lack of livelihoods, combined with migration of skilled labour and inflation may have a

negative impact on a region's economic growth. Loss of resources can lead to high costs of goods and services, delaying its development program.

Political implications: Ineffective response to relief operations during major flood events may lead to public discontent or loss of trust in the authorities or the state and national governments. Lack of development in flood-prone areas may cause social inequity and even social unrest posing threat to peace and stability in the region.

COMPONENTS REQUIRED

The **Hardware and Software Components** required are:

Table 1: Hardware Requirements

S.No.	COMPONENTS	SPECIFICATION	QUANTITY
1.	Arduino UNO	22 pins, operating voltage: 6-20V	1
2.	HC-SR04 Ultrasonic Sensor	5V, (4)I/O Pins; Operating Current: 15mA; Detection Range:50cm	1
3.	Buzzer	Operating Voltage: 4-8V; Current: 30mA	1
4.	LEDs	5mm, Operating Voltage: 5V	3 (green,orange,red)
5.	Breadboard & Jumper wires	-	1 breadboard, Wires: As per required
6.	DC Batteries and Snap Connector	Operating Voltage: 9V	1
7.	Miscellaneous	-	-
8.	LCD	16x2 Pin Configuration	1

Table 2: Software Requirements

S.No.	COMPONENTS	DESCRIPTION
1.	Arduino IDE	Arduino provides a simple and easy- to understand coding platform through the Arduino IDE. It is friendly to students and to those who are still new in electronics projects, simplifying coding and compiling and uploading codes to the board, eliminating the need for an external programmer or burner.

COMPONENTS DESCRIPTION

- **ARDUINO UNO**

The Arduino Uno is a microcontroller board which is based on the ATmega328 series controllers and has an IDE (Integrated Development Environment) for writing, compiling and uploading codes to the microcontroller. It has 14 digital input and output pins (of which 6 are PWM) and 6 analogue inputs for communication with electronic components such as sensors, switches, motors and so on. It also has 16 MHz ceramic resonators, a USB connection jack, an external power supply

jack, an ICSP (in-circuit serial programmer) header, and a reset button. Its operating voltage is 5V, input voltage 7 to 12V (limit up to 20V).

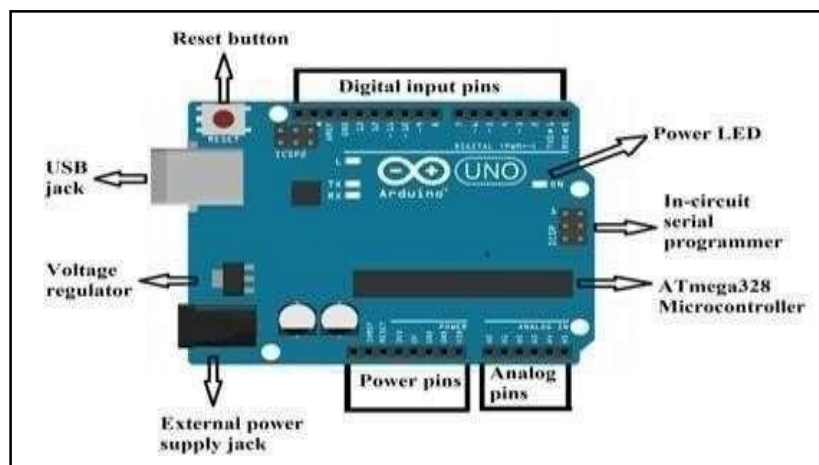


Figure 1: ARDUINO UNO

- **HC-SR04 ULTRASONIC SENSOR**

The HC-SR04 Ultrasonic Distance Sensor is a sensor used for detecting the distance to an object using sonar. It measures the distance to the target by measuring the time between the emission and reception. This sensor reads from 2cm to 400cm (0.8inch to 157inch) with an accuracy of 0.3cm (0.1inches), which is good for our project. In addition, this particular module comes with ultrasonic transmitter and receiver modules.

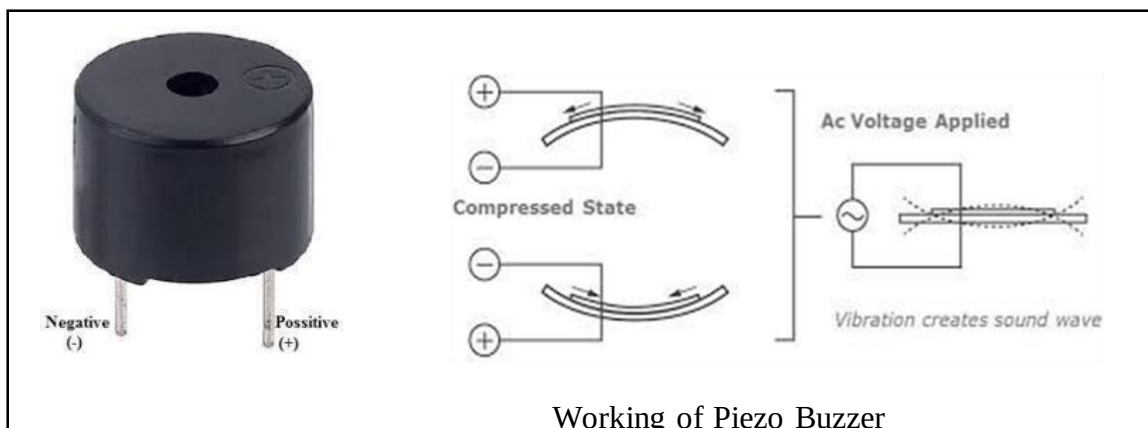


Figure 2: HC-SR04 Ultrasonic Sensor

- **BUZZER**

A "piezo buzzer" is basically a small speaker which will be connected directly to an Arduino. "Piezoelectricity" is an impression where certain crystals can deform once electricity is applied to them. By applying an electrical signal at the proper frequency, the crystal will create sound.

Figure 3: Buzzer

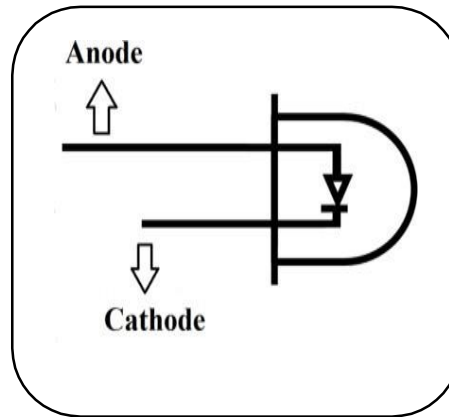


Working of Piezo Buzzer

- **LED**

A LED (Light emitting diode) is a PN junction diode which is used for emitting visible light when it is activated. When the voltage is applied over its elements, electrons regroup with holes within the LED, releasing energy in the form of photons which gives the visible light. LEDs may have dim/full capability.

Figure 4: LED

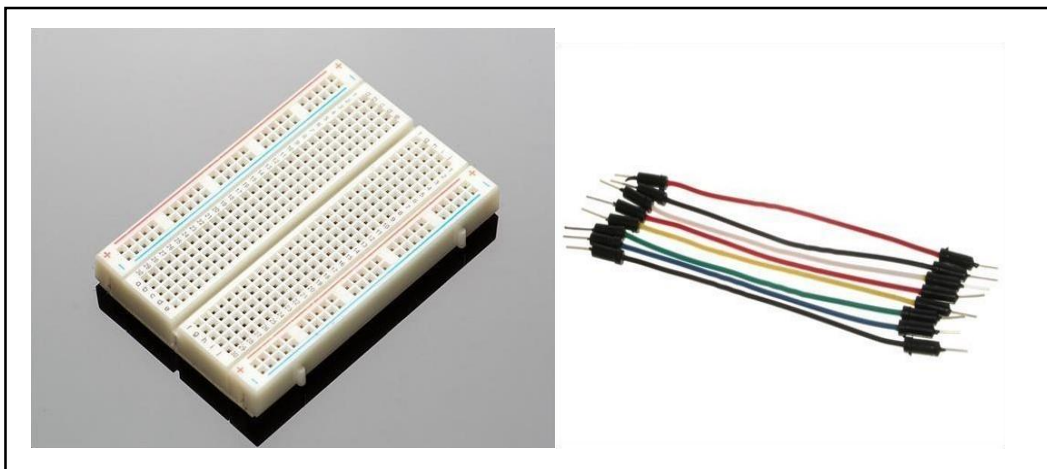


- **Breadboards and Jumper Wires**

A breadboard is a rectangular plastic board with a bunch of tinyholes in it. These holes let you easily insert electronic components to prototype an electronic circuit, with a battery, switch, resistor, and LED.

A jump wire is an electrical wire, or group of them in a cable, with a connector or pin at each end which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.

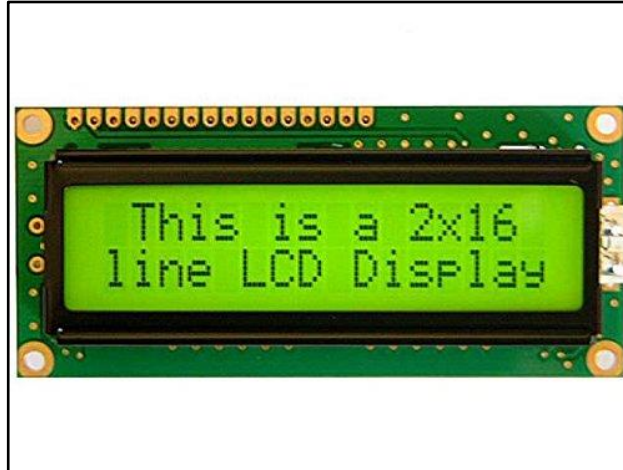
Figure 5: Breadboards and Jumper Wires



- **LCD**

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.

Figure 6: LCD



WORKING PRINCIPLE

The main working principle of this project is to detect flooding of the water bodies with ease and to be warned whenever their water level rises to a specific level. A warning signal via buzzer produces beeping sound which alerts that flooding has been occurred.

The main component used for this device is the Ultrasonic Sensor HC-SR04. The Arduino Uno microcontroller is used for communication with the ultrasonic sensor. The microcontroller sends a trigger signal to the sensor. When triggered, the sensor generates ultrasonic waves. These waves will detect the water level distance in centimeters and the waves are reflected back to the sensor. The output of the sensor is the time difference between the transmitted ultrasonic waves and the received echo signal. The microcontroller then interprets the time signal into distance. Finally, based on the water level distance, LED light of particular color will glow and also the buzzer will make a buzzing sound as an alarm when water level exceeds normal level thus detecting a possibility of flood occurring. The distance measured is displayed on the LCD.

DESIGN METHODOLOGY

CIRCUIT DIAGRAM:

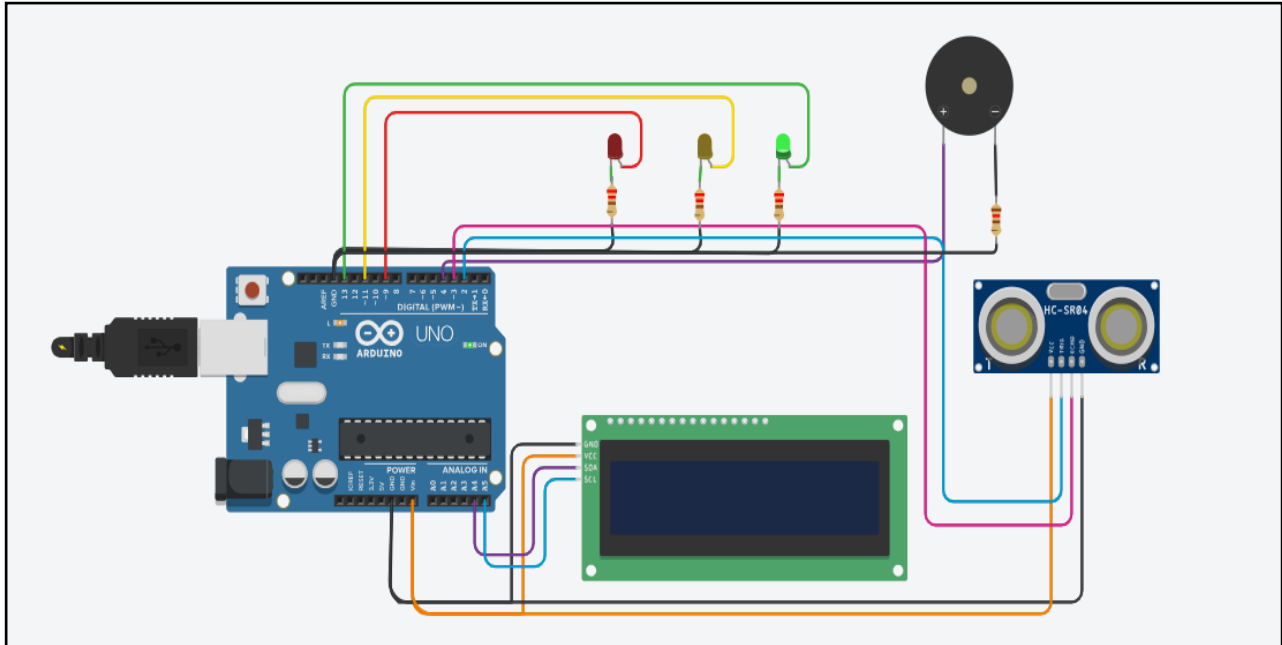


Figure 7: Circuit Diagram

CONNECTIONS:

A. LED's Connection

1) For Green LED:

- VCC of Green Color LED to Digital Pin '13' of the Arduino.
- GND of Green Color LED to the GND of Arduino via a 220ohm resistor.

2) For Orange LED:

- VCC of Orange Color LED to Digital Pin '11' of the Arduino.
- GND of Orange Color LED to the GND of Arduino via 220 ohm resistor.

3) For Red LED:

- VCC of Red Color LED to Digital Pin '9' of the Arduino.
- GND of Red Color LED to the GND of Arduino via a 220 ohm resistor.

B. Buzzer Connection

- VCC of Buzzer to Digital Pin '4' of the Arduino.
- GND of Buzzer to the GND of Arduino via 120 ohm resistor.

C. HC-SR04 Ultrasonic Sensor Connection

- VCC of Ultrasonic Sensor to Vin of Arduino.
- GND of Ultrasonic Sensor to GND of Arduino.
- Echo of Ultrasonic Sensor to Digital Pin '3' of Arduino.
- Trig of Ultrasonic Sensor to Digital Pin '2' of Arduino.

D. LCD Connection

- VCC of LCD to Vin Pin of the Arduino.
- GND of LCD to the GND of Arduino.
- SDA Pin of LCD to A4 Pin of the Arduino.
- SCL Pin of LCD to the A5 Pin of Arduino.

CIRCUIT PICTURES:

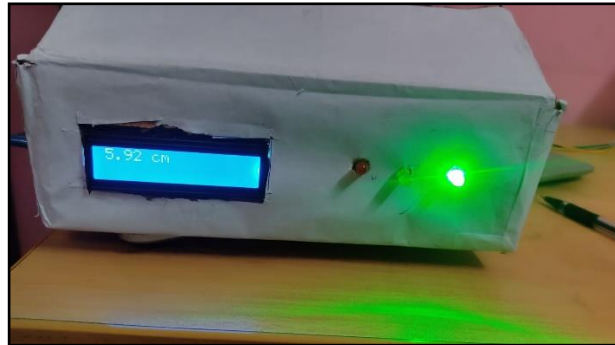


Figure 8: When water level is LOW

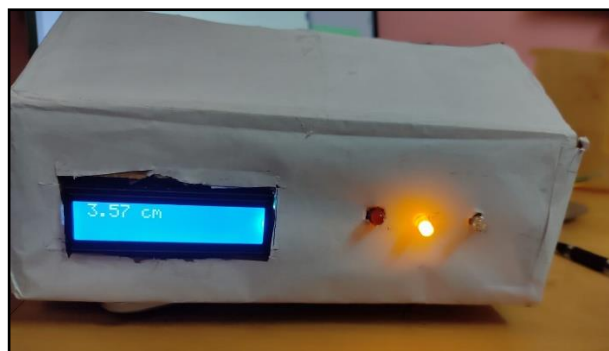


Figure 9: When water level is MODERATE



Figure 10: When water level is HIGH

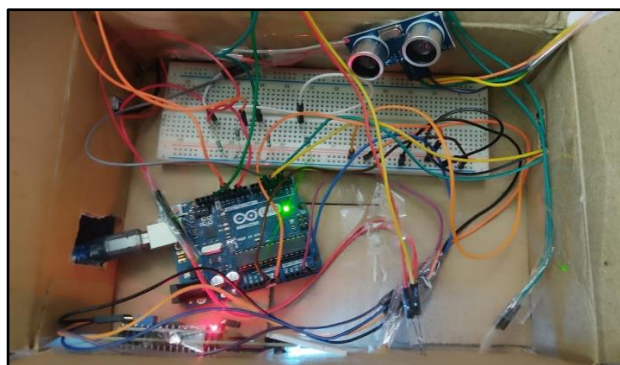


Figure 11: Circuit Picture

WORKING

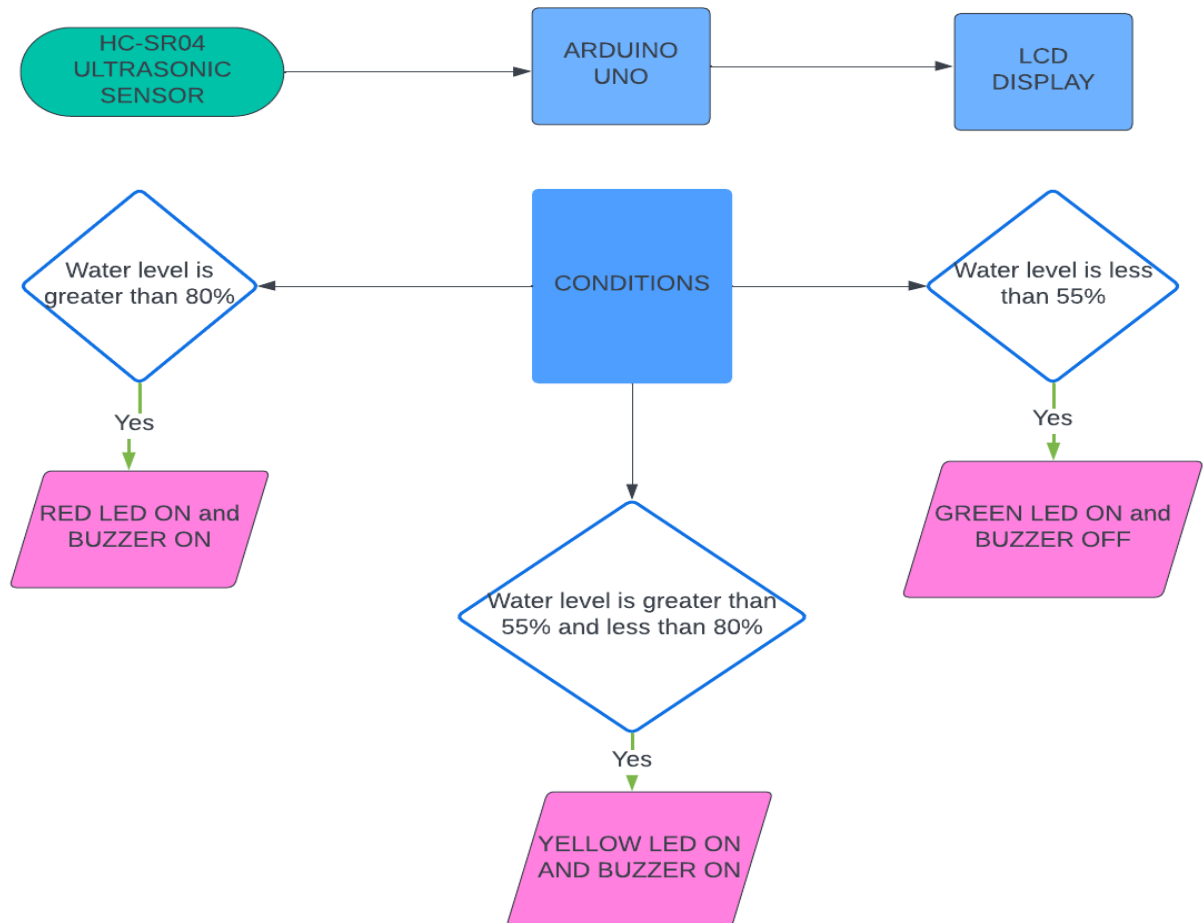


Figure 12: Workflow of the model

Firstly the USB cable type-B is connected to the Laptop's USB slot for power supply. Also the Arduino code file is run simultaneously. The ultrasonic sensor will sense the water level in distance and then the arduino program will help to convert it into centimeters. Also the distance will be displayed on LCD screen. The full water tank/container is divided into 3 zones i.e Green, Yellow and Red.

GREEN ZONE:

When water level is at Min/Normal level. That resembles 'Green Zone'. This means that water is at normal position and no sign about flood condition. Also green led will glow.

YELLOW ZONE:

When water level crosses the Intermediate level. That resembles 'Yellow Zone'. This means that water is rising and there can be chances of flood condition at that place. With increase in water level the orange led will glow and buzzer will buzz.

RED ZONE:

When water level crosses the Max Level. That resembles 'Red Zone'. This means that water level has crossed the 80% and flood situation has occurred at that place. With increase in water level the red led will glow and buzzer will buzz.

If the water level is too far, it will show a message of out of range.

WORKING CODE

Arduino Code:

//IOT Based Flood Monitoring And Alerting System.

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
// Constants for ultrasonic sensor pins
```

```
const int trigPin = 2;
```

```
const int echoPin = 3;
```

```
// Constants for LCD display
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// Constants for LED and buzzer
```

```
const int rled = 9;
```

```
const int yled = 11;
```

```
const int gled = 13;
```

```
const int buzzer = 4;
```

```
// Constants for distance thresholds
```

```
const float MAX_DISTANCE = 6.0;
```

```
const float MEDIUM_DISTANCE = 4.2;
```

```
const float MIN_DISTANCE = 2.5;
```

```
// Variables
```

```
float distance;
```

```
float smoothedDistance = 0; // Smoothed distance value
```

```
const float smoothingFactor = 0.9; // Smoothing factor for averaging
```

```

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(gled, OUTPUT);
  pinMode(yled, OUTPUT);
  pinMode(rled, OUTPUT);
  pinMode(buzzer, OUTPUT);

  lcd.init(); // Initialize the LCD display
  lcd.backlight(); // Turn on the backlight
  lcd.setCursor(0, 0);
  lcd.print("Water Level:");

  Wire.begin(); // Initialize I2C communication
  Serial.begin(9600); // Initialize Serial communication
}

void loop() {
  // Generate ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure the duration of the echo pulse
  long duration = pulseIn(echoPin, HIGH);

  // Calculate the distance in centimeters

```



```
distance = duration * 0.034 / 2.0;
```

```
// Smooth the distance readings
```

```
smoothedDistance = smoothingFactor * smoothedDistance + (1 - smoothingFactor) * distance;
```

```
// Print the smoothed distance on the LCD display
```

```
lcd.setCursor(0, 1);
```

```
lcd.clear(); // Clear the entire screen
```

```
lcd.print(smoothedDistance);
```

```
lcd.print(" cm");
```

```
delay(100);
```

```
// Check distance and activate corresponding alarms
```

```
if (smoothedDistance <= MAX_DISTANCE && smoothedDistance > MEDIUM_DISTANCE) {
```

```
    digitalWrite(gled, HIGH);
```

```
    digitalWrite(yled, LOW);
```

```
    digitalWrite(rled, LOW);
```

```
    noTone(buzzer);
```

```
    Serial.println("Water is filling up. Stay safe.");
```

```
} else if (smoothedDistance < MEDIUM_DISTANCE && smoothedDistance > MIN_DISTANCE) {
```

```
    digitalWrite(yled, HIGH);
```

```
    digitalWrite(gled, LOW);
```

```
    digitalWrite(rled, LOW);
```

```
    tone(buzzer, 1500);
```

```
    Serial.println("Water level is increasing rapidly. Take precautions.");
```

```
} else if (smoothedDistance < MIN_DISTANCE && smoothedDistance > 0) {
```

```
    digitalWrite(rled, HIGH);
```

```
    digitalWrite(gled, LOW);
```

```
digitalWrite(yled, LOW);
tone(buzzer, 2000);
Serial.println("It's already flooding. Run to safety!");
} else {
digitalWrite(gled, LOW);
digitalWrite(yled, LOW);
digitalWrite(rled, LOW);
noTone(buzzer);
Serial.println("Out of range. Cannot detect water level or flood situation.");
}

// Out of range detection
if (smoothedDistance > MAX_DISTANCE || smoothedDistance <= 0) {
Serial.println("Out of range");
noTone(buzzer);
}

delay(500);
}
```

CONCLUSION

The water level monitoring system implemented successfully utilizes an ultrasonic sensor to measure distances, translating them into actionable insights displayed on an LCD screen and signaled through LEDs and a buzzer. The system operates effectively in real-time, providing continuous monitoring of water levels within a defined range.

Key features of the system include:

Accurate Measurement: Utilizing ultrasonic technology, the system accurately measures water levels in centimeters, providing precise data for monitoring.

Visual Indicators: LEDs of different colors (red, yellow, and green) provide visual cues indicating the current water level status. This allows for easy interpretation of the situation at a glance.

Auditory Alarm: A buzzer emits distinct tones corresponding to different water level thresholds, enhancing situational awareness even in environments where visual cues might not be immediately visible.

User Interface: The LCD display presents clear information about the water level, ensuring users can quickly understand the current status without ambiguity.

Robustness: The system includes error handling mechanisms, such as detecting out-of-range measurements and displaying appropriate messages, ensuring reliability in operation.

In conclusion, the developed water level monitoring system offers an effective solution for monitoring water levels in various applications, from industrial tanks to household appliances. Its integration of multiple feedback mechanisms enhances usability and ensures timely response to changing conditions, contributing to improved water management and safety.

BIBLIOGRAPHY

- 1) Sheikh Azid, Bibhya Sharma, Krishna Raghuwaiya, Abinendra Chand, Sumeet Prasad, SMS based flood monitoring and early warning system, ARPN Journal of Engineering and Applied Science, 10(15) 2015.
- 2) Abubaker Rahmtalla, Abdalla Mohamed, Wang Guang Wei, Real Time Wireless Flood Monitoring System using Ultrasonic waves, International Journal of Science and Research (IJSR) ,3(8) 2014.
- 3) <http://www.circuitstoday.com/interface-gsm-module-with-arduino>
- 4) Manish M. Patil and Prof. Chhaya S. Khandelwal, Implementation of Patient Monitoring System Using GSM Technology. International Journal of Electronics and Communication Engineering & Technology (IJCET).4(1), 2013, pp. 18–24
- 5) S. J. Priya, S. Akshaya, E. Aruna, J. A. M. Julie, and V. Ranjani, “Flood monitoring and alerting system,” International Journal of Computer Engineering & Technology (IJCET), vol. 8, no. 2, p. 15, Mar 2017.